PathoFMPred three-representation research output

COAD TCGA-A6-A56B | TITAN, Giga-SSL and Prov-GigaPath | 2026-10-21

UNVALIDATED INTERNAL TCGA RESEARCH OUTPUT. BINARY REFERENCE RANK IS NOT PROBABILITY. DO NOT USE FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION.

Output inventory

Representation	Continuous outputs	Binary outputs	Slides pooled	Output rule
TITAN	15	16	1	Q2 >= 0.20; AUROC >= 0.60
Giga-SSL	17	13	1	Q2 >= 0.20; AUROC >= 0.60
Prov-GigaPath	18	13	1	Q2 >= 0.20; AUROC >= 0.60

Binary-call overview across the three inputs

Model	Genome doubling	APC	KRAS	PIK3CA	TP53 mutation	Any called fusion	MSI-H	Strict MSI-H	Cell Cycle	HIPPO	MYC	NOTCH	PI3K	RTK RAS	TGF-Beta	TP53 pathway
TITAN	Present	Mutation	Mutation	Mutation	Mutation	Altered	Not MSI-H	Not MSI-H	Not altered	Not altered	Not altered	Not altered	Not altered	Not altered	Not altered	Altered
Giga-SSL	Present	Mutation	Wild type	Wild type	Mutation	No output	Not MSI-H	Not MSI-H	No output	Not altered	No output	Not altered	Not altered	Not altered	Altered	Altered
Prov-GigaPath	Present	Mutation	Mutation	Wild type	Mutation	No output	Not MSI-H	Not MSI-H	No output	Not altered	No output	Not altered	Not altered	Altered	Not altered	Altered

Colour identifies an uncalibrated research class call, not the observed genotype or a probability. Exact LDA scores and matched-CV performance appear in the model tables.

Case context, not used as model input

The sigmoid-colon slide contained a moderately differentiated adenocarcinoma extending through the muscularis propria without reaching the serosal surface (pT3). The excision margins were negative, and carcinoma was present in three of 25 regional lymph nodes.

Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

How to read the report

- PathoFMPred lists representation-specific fitted objects that crossed the recorded internal performance threshold and passed the package's default sample-size rule. The inventory is not an exhaustive molecular profile.

- Continuous predictions remain in the source endpoint units. Linear predictions are unconstrained and may fall outside the source assay range. The internal TCGA rank provides display context and is not a clinical reference interval.

- Binary calls use a training-derived threshold on an uncalibrated LDA score. Neither the score nor its TCGA rank is a probability.

- The Q2, AUROC and PR-AUC values below are from the primary matched-cohort nested cross-validation; they are not five-repeat averages.

- This TCGA patient had source labels for these returned objects: TITAN 15 continuous/16 binary; Giga-SSL 17 continuous/13 binary; Prov-GigaPath 18 continuous/13 binary. Where a source label was available, the patient contributed to final full-data fitting. These predictions are not held-out case validation.

TITAN: outputs meeting internal criteria

The following tables list 15 continuous and 16 binary research outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2 (CV)	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score (Taylor)	14.870300	source endpoint units	70.1	0.296	0.572	407	effect threshold retained under grouping
MSI	MANTIS score	0.289291	source endpoint units	16.3	0.205	0.440	390	effect threshold not retained under grouping
MSI	MSIsensor score	-0.477341	source endpoint units	21.3	0.219	0.312	410	effect threshold not retained under grouping
Rna Pathway Activity	Allograft rejection	-0.469823	source endpoint units	2.7	0.241	0.494	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_INFLAMMATORY_RESPONSE	-0.430741	source endpoint units	3.4	0.202	0.473	423	effect threshold not retained under grouping
Rna Pathway Activity	Interferon-gamma response	-0.538127	source endpoint units	2.7	0.243	0.505	423	effect threshold not retained under grouping
Rna Pathway Activity	Myogenesis	-0.283684	source endpoint units	12.2	0.216	0.449	423	effect threshold not retained under grouping
Immune/context	Aneuploidy Score (Thorsson)	12.705826	source endpoint units	72.6	0.274	0.548	407	effect threshold retained under grouping
Immune/context	Indel Neoantigens	2.722714	log1p-transformed source units	26.6	0.220	0.387	322	effect threshold not retained under grouping
Immune/context	Leukocyte Fraction	0.069879	source endpoint units	2.0	0.229	0.487	418	effect threshold not retained under grouping
Immune/context	Lymphocyte Infiltration Signature Score	-0.886086	source endpoint units	7.6	0.294	0.543	416	effect threshold not retained under grouping
Immune/context	Nonsilent Mutation Rate	1.146518	log1p-transformed source units	15.8	0.257	0.312	345	effect threshold not retained under grouping
Immune/context	SNV Neoantigens	3.688697	log1p-transformed source units	11.0	0.270	0.382	350	effect threshold not retained under grouping
Immune/context	Silent Mutation Rate	0.545944	log1p-transformed source units	20.7	0.225	0.350	345	effect threshold not retained under grouping
Immune/context	TIL Regional Fraction	8.197991	source endpoint units	45.6	0.466	0.711	413	effect threshold not retained under grouping

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Present	0.686	67.0	0.752	0.668	407 (165/242)	effect threshold retained under grouping
Mutation	APC	Mutation	2.111	73.2	0.838	0.792	382 (196/186)	effect threshold retained under grouping
Mutation	KRAS	Mutation	-0.918	47.5	0.776	0.588	382 (126/256)	effect threshold retained under grouping
Mutation	PIK3CA	Mutation	-1.437	49.3	0.637	0.287	382 (83/299)	effect threshold not retained under grouping
Mutation	TP53	Mutation	1.158	73.2	0.859	0.811	382 (161/221)	effect threshold retained under grouping
Fusion	Any called fusion	Altered	0.698	67.9	0.619	0.638	257 (144/113)	effect threshold not retained under grouping
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-3.729	42.7	0.823	0.601	390 (73/317)	effect threshold retained under grouping

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-7.494	46.5	0.900	0.737	373 (56/317)	effect threshold retained under grouping
Pathway	Cell Cycle	Not altered	-0.213	28.3	0.615	0.696	253 (147/106)	effect threshold not retained under grouping
Pathway	HIPPO	Not altered	-4.000	21.8	0.675	0.372	323 (63/260)	effect threshold retained under grouping
Pathway	MYC	Not altered	-4.726	2.0	0.614	0.344	323 (79/244)	effect threshold not retained under grouping
Pathway	NOTCH	Not altered	-1.457	38.9	0.628	0.445	323 (107/216)	effect threshold not retained under grouping
Pathway	PI3K	Not altered	-0.978	33.0	0.604	0.529	323 (148/175)	effect threshold retained under grouping
Pathway	RTK RAS	Not altered	-0.279	10.7	0.678	0.858	323 (242/81)	effect threshold retained under grouping
Pathway	TGF-Beta	Not altered	-2.289	12.5	0.620	0.398	323 (100/223)	effect threshold retained under grouping
Pathway	TP53	Altered	1.985	81.3	0.766	0.843	323 (208/115)	effect threshold retained under grouping

Giga-SSL: outputs meeting internal criteria

The following tables list 17 continuous and 13 binary research outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2 (CV)	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score (Taylor)	18.139590	source endpoint units	90.3	0.264	0.517	407	effect threshold not retained under grouping
MSI	MANTIS score	0.292815	source endpoint units	15.3	0.240	0.424	390	effect threshold not retained under grouping
MSI	MSIsensor score	-1.894260	source endpoint units	9.9	0.251	0.325	410	effect threshold retained under grouping
Rna Pathway Activity	Allograft rejection	-0.345540	source endpoint units	11.0	0.337	0.596	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_ANGIOGENESIS	-0.302477	source endpoint units	10.5	0.202	0.458	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_APICAL_JUNCTION	-0.141270	source endpoint units	23.0	0.209	0.456	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_COMPLEMENT	-0.212343	source endpoint units	16.0	0.270	0.554	423	effect threshold retained under grouping
Rna Pathway Activity	IL6/JAK/STAT3	-0.248587	source endpoint units	19.3	0.271	0.539	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_INFLAMMATORY_RESPONSE	-0.220755	source endpoint units	20.0	0.234	0.532	423	effect threshold not retained under grouping
Rna Pathway Activity	Interferon-alpha response	-0.501981	source endpoint units	3.0	0.265	0.480	423	effect threshold not retained under grouping
Rna Pathway Activity	Interferon-gamma response	-0.444470	source endpoint units	7.7	0.304	0.563	423	effect threshold not retained under grouping
Rna Pathway Activity	Myogenesis	-0.116515	source endpoint units	30.9	0.221	0.449	423	effect threshold not retained under grouping
Immune/context	Aneuploidy Score (Thorsson)	15.280484	source endpoint units	90.8	0.227	0.493	407	effect threshold not retained under grouping
Immune/context	Leukocyte Fraction	0.081553	source endpoint units	6.8	0.301	0.565	418	effect threshold retained under grouping
Immune/context	Lymphocyte Infiltration Signature Score	-0.334818	source endpoint units	30.2	0.292	0.534	416	effect threshold not retained under grouping
Immune/context	Macrophage Regulation	-0.763750	source endpoint units	14.5	0.308	0.556	416	effect threshold not retained under grouping
Immune/context	TIL Regional Fraction	18.020547	source endpoint units	92.4	0.399	0.632	413	effect threshold not retained under grouping

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Present	1.527	87.6	0.770	0.686	407 (165/242)	effect threshold retained under grouping
Mutation	APC	Mutation	1.845	64.3	0.820	0.780	382 (196/186)	effect threshold not retained under grouping
Mutation	KRAS	Wild type	-1.862	29.7	0.729	0.509	382 (126/256)	effect threshold not retained under grouping
Mutation	PIK3CA	Wild type	-2.362	33.4	0.666	0.337	382 (83/299)	effect threshold not retained under grouping

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Mutation	TP53	Mutation	0.231	55.4	0.780	0.673	382 (161/221)	effect threshold not retained under grouping
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-5.984	25.0	0.788	0.594	390 (73/317)	effect threshold retained under grouping
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-9.182	33.1	0.846	0.612	373 (56/317)	effect threshold retained under grouping
Pathway	HIPPO	Not altered	-3.218	54.6	0.616	0.337	323 (63/260)	effect threshold not retained under grouping
Pathway	NOTCH	Not altered	-1.324	33.3	0.625	0.474	323 (107/216)	effect threshold retained under grouping
Pathway	PI3K	Not altered	-1.018	33.9	0.637	0.571	323 (148/175)	effect threshold retained under grouping
Pathway	RTK RAS	Not altered	0.384	28.9	0.643	0.859	323 (242/81)	effect threshold retained under grouping
Pathway	TGF-Beta	Altered	-0.872	58.4	0.607	0.376	323 (100/223)	effect threshold not retained under grouping
Pathway	TP53	Altered	2.481	83.1	0.737	0.843	323 (208/115)	effect threshold retained under grouping

Prov-GigaPath: outputs meeting internal criteria

The following tables list 18 continuous and 13 binary research outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2 (CV)	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score (Taylor)	17.678433	source endpoint units	88.1	0.306	0.576	407	effect threshold retained under grouping
MSI	MANTIS score	0.289534	source endpoint units	14.2	0.292	0.452	390	effect threshold not retained under grouping
MSI	MSIsensor score	-2.185303	source endpoint units	10.6	0.300	0.352	410	effect threshold not retained under grouping
Rna Pathway Activity	Allograft rejection	-0.504984	source endpoint units	7.9	0.339	0.604	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_COMPLEMENT	-0.385858	source endpoint units	4.4	0.251	0.551	423	effect threshold retained under grouping
Rna Pathway Activity	HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	-0.359894	source endpoint units	15.5	0.201	0.471	423	effect threshold not retained under grouping
Rna Pathway Activity	IL6/JAK/STAT3	-0.512007	source endpoint units	2.7	0.224	0.501	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_INFLAMMATORY_RESPONSE	-0.477819	source endpoint units	5.3	0.254	0.547	423	effect threshold not retained under grouping
Rna Pathway Activity	Interferon-alpha response	-0.832801	source endpoint units	0.1	0.275	0.515	423	effect threshold retained under grouping
Rna Pathway Activity	Interferon-gamma response	-0.709889	source endpoint units	0.8	0.332	0.586	423	effect threshold retained under grouping
Immune/context	Aneuploidy Score (Thorsson)	14.624235	source endpoint units	86.4	0.274	0.544	407	effect threshold retained under grouping
Immune/context	Fraction Altered	0.706127	source endpoint units	96.2	0.217	0.506	413	effect threshold not retained under grouping
Immune/context	Leukocyte Fraction	0.093382	source endpoint units	10.2	0.259	0.533	418	effect threshold retained under grouping
Immune/context	Lymphocyte Infiltration Signature Score	-0.984381	source endpoint units	6.4	0.275	0.549	416	effect threshold not retained under grouping
Immune/context	Macrophage Regulation	-0.783203	source endpoint units	17.9	0.334	0.582	416	effect threshold not retained under grouping
Immune/context	SNV Neoantigens	3.449005	log1p-transformed source units	4.4	0.242	0.327	350	effect threshold retained under grouping
Immune/context	TIL Regional Fraction	7.605708	source endpoint units	47.1	0.508	0.688	413	effect threshold not retained under grouping
Immune/context	Th1 Cells	-1581.304130	source endpoint units	1.6	0.240	0.513	416	effect threshold not retained under grouping

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Present	2.120	87.3	0.779	0.712	407 (165/242)	effect threshold retained under grouping
Mutation	APC	Mutation	5.028	86.5	0.822	0.775	382 (196/186)	effect threshold retained under grouping
Mutation	KRAS	Mutation	-0.355	61.9	0.746	0.538	382 (126/256)	effect threshold not retained under grouping
Mutation	PIK3CA	Wild type	-4.653	19.0	0.641	0.294	382 (83/299)	effect threshold not retained under grouping

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Mutation	TP53	Mutation	3.277	95.7	0.797	0.688	382 (161/221)	effect threshold not retained under grouping
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-5.655	37.8	0.793	0.587	390 (73/317)	effect threshold retained under grouping
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-9.665	21.6	0.909	0.703	373 (56/317)	effect threshold retained under grouping
Pathway	HIPPO	Not altered	-4.563	22.4	0.632	0.308	323 (63/260)	effect threshold retained under grouping
Pathway	NOTCH	Not altered	-1.467	45.4	0.638	0.454	323 (107/216)	effect threshold retained under grouping
Pathway	PI3K	Not altered	-2.160	15.6	0.624	0.576	323 (148/175)	effect threshold retained under grouping
Pathway	RTK RAS	Altered	4.349	72.9	0.702	0.857	323 (242/81)	effect threshold retained under grouping
Pathway	TGF-Beta	Not altered	-3.030	21.5	0.632	0.424	323 (100/223)	effect threshold retained under grouping
Pathway	TP53	Altered	4.208	83.7	0.710	0.791	323 (208/115)	effect threshold retained under grouping

Reference labels and performance estimates

Q2, AUROC and PR-AUC in these tables are primary matched-cohort nested-CV estimates, not means of five repeats. The patient-level values and class calls come from final models fitted to all eligible labelled TCGA patients. The example patients contributed to fitting a final object whenever its reference label is available; these case predictions are not held-out validation. Grouped sensitivity was evaluated for all three representations by separating TCGA tissue-source-site codes in both outer and inner folds. The code is not a scanner, laboratory or independent institution identifier.

Feature group	Reference label source
Mutations	TCGA MC3 protein-altering calls within the Bailey tissue-specific driver catalogue
Aneuploidy score (Taylor)	Arm-level gain/loss burden from Taylor et al., Cancer Cell 2018, Table S2 (doi:10.1016/j.ccell.2018.03.007)
Aneuploidy Score (Thorsson)	Separate genomic-context score distributed in the PanImmune matrix of Thorsson et al., Immunity 2018 (doi:10.1016/j.immuni.2018.03.023)
Genome doubling	Binary copy-number call from Taylor et al., Cancer Cell 2018, Table S2
MSI	cBioPortal TCGA PanCancer MANTIS and MSIsensor variables
Oncogenic pathways	Sanchez-Vega pathway alteration matrix
Immune and genomic context	Published Thorsson PanImmune computational phenotypes